SnapOpt

HEN Superstructure Optimization from Excel — User Guide

User Guide

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SnapTera: From Targeting to Optimization to Network Design

SnapOpt — Optimization Results

Run 'test-excel-frontend' | solver dicopt | 2026-07-10 18:30 | C:\Users\DELL\Desktop\College\SEMESTE

Solve & Cost Summary

Quantity	Value
QH,min (kW)	2,509.874
QC,min (kW)	961.372
Q recovered (kW)	17,106.173
Total area (m²)	2,572.286
Number of units	14.0
Heating cost ($/yr)	322,518.855
Cooling cost ($/yr)	6,177.774
OPEX ($/yr)	328,696.629
ACAPEX ($/yr)	46,765.302
CAPEX ($)	287,352.539
TAC ($/yr)	375,461.931
CRF (-)	0.163
Solver status	Normal Completion
Model status	8 Integer Solution
Objective ($/yr)	375,461.931

Matches (process-to-process exchangers)

Hot	Cold	Stage	Duty (kW)	Area (m²)
H1	C4	1	30.106	0.764
H1	CMIX	2	6,489.264	401.221
H2	C3	3	1,969.849	156.303
H3	CMIX	3	0.0	0.0
H1	C3	4	3,028.123	748.2
H2	CMIX	4	2,258.909	415.342
H3	C3	4	1,753.483	409.734
H3	CMIX	4	913.667	163.451
H1	C3	5	363.471	156.669
H2	C3	5	299.302	120.603

Utility Exchangers

Stream	Kind	Duty (kW)	Area (m²)	Approach (°C)
H1	cooler	225.891	31.642	11.0
H2	cooler	170.87	23.935	11.0
H3	cooler	564.611	47.819	25.013
CMIX	heater	2,509.874	195.672	24.138

Stage-Boundary Temperatures (°C)

Stream	Side	T loc 1	T loc 2	T loc 3
H1	hot	397.0	395.9	166.1
H2	hot	258.0	258.0	258.0
H3	hot	164.0	164.0	164.0
CMIX	cold	325.9	325.9	141.9
C3	cold	182.0	182.0	182.0
C4	cold	263.0	258.0	258.0

SnapOpt finds the cost-optimal heat exchanger network with the SSGO stagewise superstructure model (Yee & Grossmann, 1990; Chen LMTD areas), solved rigorously in GAMS — DICOPT for a fast local optimum, SCIP for a global search. From Excel, one click on Optimize runs the whole chain: the Streams sheet is exported, the model is generated and solved, and the optimized matches, duties, areas and cost breakdown come back as a live results sheet.

1. Requirements and installation

Component	Details
Microsoft Excel (Windows)	Hosts the front-end add-in SnapOpt.xlam; macros must be enabled.
Python 3	Runs the SnapOpt engine (snapopt_run.py / snapopt_import.py); must be on PATH (or set the 'Python command' box).
GAMS + solvers	GAMS with DICOPT, SCIP, CPLEX and CONOPT licenses; auto-detected on PATH or in C:\GAMS\<version>\.
SnapOpt folder	The folder containing snapopt_run.py. The template pre-fills it; fix the 'SnapOpt folder' box if you move things.

Install the add-in: double-click `SnapOpt.xlam` for one session, or Excel → File → Options → Add-ins → Manage: Excel Add-ins → Go… → Browse… → select the file. If Windows blocks the download, right-click the file → Properties → Unblock first.

2. Workflow

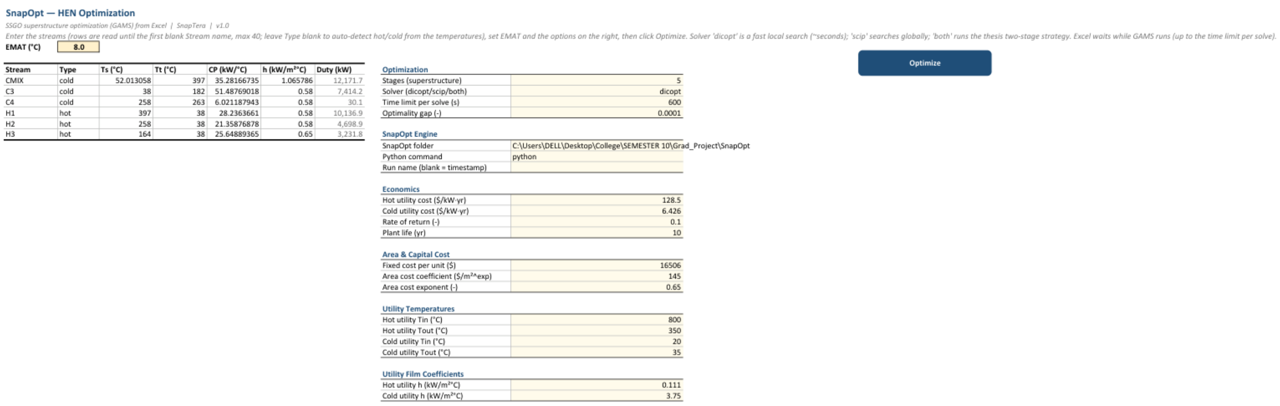
Step	Ribbon button	What happens
1	New Template	Streams input workbook, pre-filled with the kerosene hydrotreater demo case (H1–H3, CMIX, C3, C4 + economics), plus a Credits sheet.
2	—	Enter your streams (rows are read until the first blank Stream name, max 40; a blank Type is auto-detected from the temperatures), set EMAT (B4) and the option boxes.
3	Optimize	Streams → input.json → SSGO model generated → GAMS solves → results imported into the 'SnapOpt Results' sheet. Excel waits while GAMS runs.

Solver modes

Solver	What it does	Typical time (demo case)
`dicopt`	Fast local search (DICOPT MINLP, CPLEX + CONOPT subsolvers). Good first answer.	~1–2 s
`scip`	Global search — keeps improving until the time limit or the optimality gap.	up to the time limit
`both`	The thesis two-stage strategy: DICOPT warm start, then SCIP global refinement. Never worse than dicopt alone.	dicopt + time limit

On the kerosene demo, both with a 60 s time limit lands within $0.02/yr of the thesis 10,000 s global optimum (TAC $364,076.70/yr).

3. The Streams input sheet


SnapOpt — HEN Optimization

SSGO superstructure optimization (GAMS) from Excel | SnapTera | v1.0

Enter the streams (rows are read until the first blank Stream name, max 40; leave Type blank to auto-detect hot/cold from the temperatures), set EMAT and the options on the right, then click Optimize. Solver 'dicopt' is a fast local search (~seconds); 'scip' searches globally; 'both' runs the thesis two-stage strategy. Excel waits while GAMS runs (up to the time limit per solve).

EMAT (°C) 8.0

Stream	Type	Ts (°C)	Tt (°C)	CP (kW/°C)	h (kW/m²°C)	Duty (kW)
CMIX	cold	52.013058	397	35.28166735	1.065786	12,171.7
C3	cold	38	182	51.48769018	0.58	7,414.2
C4	cold	258	263	6.021187943	0.58	30.1
H1	hot	397	38	28.2363661	0.58	10,136.9
H2	hot	258	38	21.35876878	0.58	4,698.9
H3	hot	164	38	25.64889365	0.65	3,231.8

Optimization

Stages (superstructure)	5
Solver (dicopt/scip/both)	dicopt
Time limit per solve (s)	600
Optimality gap (-)	0.0001

SnapOpt Engine

SnapOpt folder	C:\Users\DELL\Desktop\College\SEMESTER 10\Grad_Project\SnapOpt
Python command	python
Run name (blank = timestamp)	

Economics

Hot utility cost ($/kW·yr)	128.5
Cold utility cost ($/kW·yr)	6.426
Rate of return (-)	0.1
Plant life (yr)	10

Area & Capital Cost

Fixed cost per unit ($)	16506
Area cost coefficient ($/m²^exp)	145
Area cost exponent (-)	0.65

Utility Temperatures

Hot utility Tin (°C)	800
Hot utility Tout (°C)	350
Cold utility Tin (°C)	20
Cold utility Tout (°C)	35

Utility Film Coefficients

Hot utility h (kW/m²°C)	0.111
Cold utility h (kW/m²°C)	3.75

Optimize

The input template: EMAT box, self-reading stream table with live Duty column (left), option boxes (right), Optimize button.

Input box	Fields
EMAT (B4)	Exchanger minimum approach temperature for the optimization (°C).

Input box	Fields
Optimization	Stages (superstructure depth, default 5) • Solver (dicopt/scip/both) • Time limit per solve (s) • Optimality gap.
SnapOpt Engine	SnapOpt folder • Python command • Run name (each run gets its own runs\<name>\ workspace; blank = timestamp).
Economics	Hot/cold utility cost ($/kW·yr), rate of return, plant life.
Area & Capital Cost	Fixed cost per unit ($), area cost coefficient, area cost exponent.
Utility Temperatures / Film Coefficients	Hot and cold utility inlet/outlet temperatures and film coefficients.

4. The results sheet

SnapOpt — Optimization Results

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'SnapOpt Results' after a DICOPT run of the demo case.

Section	Contents
Solve & Cost Summary	QH,min / QC,min / Q recovered, total area, number of units, heating/cooling cost, OPEX, ACAPEX, CAPEX, TAC, CRF, solver and model status.
Matches	Every process-to-process exchanger: hot and cold stream, superstructure stage, duty (kW) and Chen-approximation area (m²).

Section	Contents
Utility Exchangers	Coolers and heaters with duty, area and approach temperature.
Stage-Boundary Temperatures	Each stream's temperature at every superstructure location — the data SnapNet uses to draw the network.

Every run keeps its full workspace in runs\<name>\: input.json, the generated model.gms, the GAMS listing, and results*.csv + solution.json — the hand-off file SnapNet draws from.

5. Command line and Python use

The Excel button is a thin front-end over the engine, which you can run directly:

Command	Effect
`python snapopt_run.py`	Optimize the built-in kerosene demo with DICOPT.
`python snapopt_run.py case.json --solver both --reslim 3600`	Optimize a JSON-defined case with the two-stage strategy, 1 h per solve.
`python snapopt_run.py --dry-run`	Generate model.gms without solving (inspection).
`python snapopt_import.py path\to\model.lst`	Re-import any solved SSGO listing into CSV/JSON exports.

The input JSON (see `example_input.json`) lists streams as {name, ts, tt, cp, h, [type]} plus optional dtmin, stages and economics overrides. From Python: `from snapopt_run import run_ssgo`.

6. Model and validation

Model: SSGO — the stagewise superstructure MINLP of Yee & Grossmann (1990): isothermal-mixing stages, binary match variables, Chen LMTD approximation in the area cost terms, simultaneous energy–area–units trade-off in one TAC objective. The generated model reproduces the graduate-thesis model GP17.99 equation-for-equation and is stream-count-generic.

Validation: the importer is oracle-tested against hand-transcribed thesis values; a live DICOPT run of the generated demo model reproduces the thesis listing exactly (TAC $375,461.93/yr, 10 matches, 0.0 kW duty deviation); the Excel harness re-checks the whole chain end-to-end on every build. The SSGO results were cross-validated against Aspen Energy Analyzer and pinch supertargeting in the parent thesis.

SnapOpt v1.0 — created by Mohammed Ali Mansoury, 2026. MIT License. Part of the SnapTera suite with SnapTarget (targeting) and SnapNet (network drawing). Contact: atsnaptera@gmail.com